MODULE 4: JAVABEAN

GitHub Repository Link: https://github.com/devcnx/csd430

styles.css

```
/*
    styles.css - External CSS for Module 4 assignment
    Author: Brittaney Perry-Morgan
    Description: Styles for the JavaBean movie data display.
 */

/* ==========================
   1. GLOBAL STYLES
   ========================== */

*,
html,
body {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: 'Segoe UI', Arial, sans-serif;
    background: linear-gradient(120deg, #f8fafc 0%, #e0e7ef 100%);
    margin: 0;
    padding: 20px;
    min-height: 100vh;
}

/* ==========================
   2. LAYOUT & CONTAINER
   ========================== */

.container,
.top-title {
    max-width: 900px;
    margin: 0 auto;
    padding: 20px;
}
```

```
.container {
    background: #fff;
    border-radius: 12px;
    box-shadow: 0 4px 24px rgba(0, 0, 0, 0.08);
    margin-bottom: 20px;
}

.top-title {
    text-align: center;
    background: linear-gradient(135deg, #2a3a5e 0%, #1a2540 100%);
    color: #fff;
}

.top-title h1 {
    color: #fff;
    margin: 0 0 10px 0;
}

.top-title h2 {
    color: #e0e7ef;
    margin: 0;
    font-weight: 400;
}

.top-title h2 em {
    font-style: normal;
    color: #a3d9ff;
}

/* ========================
   3. TYPOGRAPHY
   ======================== */

h1,
h2,
h3 {
    color: #2a3a5e;
    margin: 20px 0 15px 0;
    text-transform: uppercase;
}
```

```
h1 {
    font-weight: 200;
    font-size: 1.8rem;
}

h2 {
    font-weight: 400;
    font-size: 1.4rem;
}

h3 {
    font-weight: 600;
    font-size: 1.1rem;
    color: #1a2540;
    border-bottom: 2px solid #e3eaf6;
    padding-bottom: 8px;
}

p.description {
    color: #ffffff;
    line-height: 1.7;
    margin: 15px auto;
    padding: 0 20px;
}

/* ========================
   4. LISTS
   ======================== */

.field-descriptions {
    list-style: none;
    padding: 0 20px;
}

.field-descriptions li {
    padding: 10px 15px;
    margin: 8px 0;
    background: #f1f5fa;
    border-radius: 6px;
    border-left: 4px solid #2a3a5e;
    color: #444;
```

```
}

/* ========================
   5. FORMS
   ========================= */

.form-table {
    width: 100%;
    border-collapse: collapse;
    margin: 20px;
    background: #fff;
}

.form-table tr {
    background: #fff;
}

.form-table td {
    border: 1px solid #e3eaf6;
    padding: 12px 15px;
    vertical-align: top;
}

.form-table td:first-child {
    background: #f9fbfd;
    width: 25%;
    font-weight: 600;
    color: #1a2540;
}

.form-table td:last-child {
    width: 75%;
}

.form-table label {
    display: block;
    margin-bottom: 5px;
    font-size: 0.9rem;
    color: #333;
}
```

```
.form-table input[type="text"],
.form-table input[type="email"],
.form-table input[type="tel"],
.form-table input[type="number"],
.form-table input[type="date"],
.form-table select,
.form-table textarea {
    width: 100%;
    padding: 10px 12px;
    border: 1px solid #ccc;
    border-radius: 6px;
    font-size: 1rem;
    font-family: 'Segoe UI', Arial, sans-serif;
    box-sizing: border-box;
}

.form-table select {
    background: #fff;
    cursor: pointer;
}

.form-table textarea {
    resize: vertical;
    min-height: 80px;
}

.form-table input[type="checkbox"] {
    width: auto;
    margin-right: 8px;
}

/* ========================
   6. NAVIGATION
   ======================== */

nav ul {
    list-style: none;
    padding: 0 20px;
}

nav ul li {
```

```
    margin: 10px 0;
}

nav ul li a {
    color: #2a3a5e;
    text-decoration: none;
    font-weight: 600;
    padding: 8px 12px;
    background: #f1f5fa;
    border-radius: 6px;
    transition: all 0.2s ease;
}

nav ul li a:hover {
    background: #2a3a5e;
    color: #fff;
}

/* ========================
   6. FORM ACTIONS (for button links)
   ======================== */

.form-actions {
    margin: 30px 20px;
    padding: 15px;
    background: #f9fbfd;
    border-radius: 8px;
    text-align: center;
}

.btn-submit,
.btn-reset {
    padding: 12px 24px;
    margin: 0 10px;
    font-size: 1rem;
    font-weight: 600;
    border: none;
    border-radius: 6px;
    cursor: pointer;
    transition: all 0.2s ease;
    text-decoration: none;
```

```
    display: inline-block;
}

.btn-submit {
    background: #2a3a5e;
    color: #fff;
}

.btn-submit:hover {
    background: #1a2540;
    transform: translateY(-1px);
    box-shadow: 0 4px 12px rgba(42, 58, 94, 0.3);
}

.btn-reset {
    background: #e3eaf6;
    color: #2a3a5e;
}

.btn-reset:hover {
    background: #cdd8eb;
}

/* ========================
   7. FORM ACTIONS (for button links)
   ========================= */

.form-actions {
    margin: 30px 20px;
    padding: 15px;
    background: #f9fbfd;
    border-radius: 8px;
    text-align: center;
}

.btn-submit,
.btn-reset {
    padding: 12px 24px;
    margin: 0 10px;
    font-size: 1rem;
    font-weight: 600;
```

```
    border: none;
    border-radius: 6px;
    cursor: pointer;
    transition: all 0.2s ease;
    text-decoration: none;
    display: inline-block;
}

.btn-submit {
    background: #2a3a5e;
    color: #fff;
}

.btn-submit:hover {
    background: #1a2540;
    transform: translateY(-1px);
    box-shadow: 0 4px 12px rgba(42, 58, 94, 0.3);
}

.btn-reset {
    background: #e3eaf6;
    color: #2a3a5e;
}

.btn-reset:hover {
    background: #cdd8eb;
}

/* ========================
   8. TABLES
   ======================== */

table {
    width: 100%;
    border-collapse: collapse;
    margin: 20px;
    background: #fff;
}

thead {
    background: #2a3a5e;
```

```
    color: #fff;
}

th,
td {
    border: 1px solid #e3eaf6;
    padding: 12px 15px;
    text-align: left;
}

th {
    font-weight: 600;
    font-size: 0.85rem;
    text-transform: uppercase;
    letter-spacing: 0.5px;
}

td {
    font-size: 0.95rem;
    color: #333;
    vertical-align: top;
}

tr:nth-child(even) {
    background: #f9fbfd;
}

tr:hover {
    background: #e8f0ff;
    transition: background 0.2s ease;
}

/* ========================
   9. FOOTER
   ======================== */

footer {
    text-align: center;
    padding: 20px;
    color: #666;
    font-size: 0.9rem;
```

```
}

footer p {
    margin: 0;
}

/* ========================
   10. RESPONSIVE DESIGN
   ======================== */

@media (max-width: 768px) {
    body {
        padding: 10px;
    }

    .container,
    .top-title {
        padding: 15px;
    }

    h1 {
        font-size: 1.4rem;
    }

    h2 {
        font-size: 1.1rem;
    }

    h3 {
        font-size: 1rem;
    }

    .form-table {
        margin: 10px 0;
        font-size: 0.85rem;
    }

    .form-table td {
        padding: 8px 10px;
    }
```

```
.form-table td:first-child {
    font-size: 0.85rem;
}

.form-table input[type="text"],
.form-table input[type="email"],
.form-table input[type="tel"],
.form-table input[type="number"],
.form-table input[type="date"],
.form-table select,
.form-table textarea {
    font-size: 0.9rem;
    padding: 8px;
}

table {
    margin: 10px 0;
    font-size: 0.85rem;
}

th,
td {
    padding: 8px 10px;
}

p.description {
    padding: 0 10px;
}

.field-descriptions {
    padding: 0 10px;
}

.field-descriptions li {
    padding: 8px 12px;
}

nav ul {
    padding: 0 10px;
}
```

```
    .form-actions {
        margin: 20px 10px;
        padding: 10px;
    }

    .btn-submit,
    .btn-reset {
        padding: 10px 16px;
        font-size: 0.9rem;
        margin: 5px;
    }
}
```

index.jsp

```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<%--
    CSD430 Server-Side Development

    Module 4: JavaBean
    Author: Brittaney Perry-Morgan

    Description: Main entry point for Module 4. This page demonstrates
    the JavaBean by accessing movie data and displaying it using JSP
    scriptlets with external CSS styling.

    index.jsp - Module 4 Home Page
--%>
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Module 4 Assignment: JavaBean</title>
    <link rel="stylesheet" type="text/css" href="css/styles.css">
</head>
<body>
    <div class="container top-title">
        <h1>Module 4 Assignment: JavaBean</h1>
        <h2>Movie: <em>Subservience</em> (2024)</h2>
        <p class="description">This web application demonstrates JavaBean usage for storing and displaying movie data. The
MovieBean class holds all movie information and is accessed via JSP scriptlets.</p>
    </div>
```

```
  <div class="container">
    <h3>Module Overview</h3>
    <ul class="field-descriptions">
      <li><strong>JavaBean:</strong> MovieBean - A Serializable Java class that encapsulates all movie data</li>
      <li><strong>Data Categories:</strong> Cast Details, Production Information, Technical Specifications, Plot Elements,
Release Information</li>
      <li><strong>Data Points:</strong> 24 total records across 5 categories with 3 fields each</li>
      <li><strong>Display Method:</strong> JSP scriptlets access JavaBean properties and render HTML tables</li>
    </ul>

    <h3>Navigation</h3>
    <nav>
      <ul>
        <li><a href="movieDisplay.jsp">View Movie Data</a> - Display all movie information from the JavaBean</li>
        <li><a href="castDisplay.jsp">View Cast Details</a> - Display only the cast information</li>
        <li><a href="productionDisplay.jsp">View Production Info</a> - Display production details</li>
      </ul>
    </nav>

    <h3>JavaBean Features</h3>
    <table>
      <thead>
        <tr>
          <th>Feature</th>
          <th>Description</th>
        </tr>
      </thead>
      <tbody>
        <tr>
          <td>Serializable</td>
          <td>Implements java.io.Serializable for session storage</td>
        </tr>
        <tr>
          <td>Default Constructor</td>
          <td>Initializes all movie data upon instantiation</td>
        </tr>
        <tr>
          <td>Getters/Setters</td>
          <td>Public accessor methods for all private fields</td>
        </tr>
        <tr>
```

```
                <td>Data Encapsulation</td>
                <td>All movie data stored as private String arrays</td>
            </tr>
            <tr>
                <td>Fields</td>
                <td>5 data categories with 24 total records</td>
            </tr>
        </tbody>
    </table>
  </div>

  <footer>
    <p>&copy; 2026 Brittaney Perry-Morgan | CSD430 Server-Side Development</p>
  </footer>
</body>
</html>
```

castDisplay.jsp

```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<%@ page import="com.example.MovieBean" %>
<%--
  CSD430 Server-Side Development

  Module 4: JavaBean
  Author: Brittaney Perry-Morgan

  Description: Displays only the Cast Details from the MovieBean JavaBean.
  All Java code is within Scriptlets. All HTML tags are outside of
  Scriptlets. External CSS is used for styling.

  castDisplay.jsp - Cast Details Display Page
--%>

<%-- ========================
   1. JAVA BEAN INSTANTIATION
   ========================= --%>
<%
  // Create a new MovieBean instance
  MovieBean movie = new MovieBean();
%>
```

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Module 4 Assignment: JavaBean - Cast Details</title>
    <link rel="stylesheet" type="text/css" href="css/styles.css">
</head>
<body>
    <div class="container top-title">
        <h1>Module 4 Assignment: JavaBean</h1>
        <h2>Movie: <em><%= movie.getMovieTitle() %></em> (<%= movie.getMovieYear() %>)</h2>
        <p class="description">This page displays the cast details for the movie, retrieved from the MovieBean JavaBean.</p>
    </div>

    <div class="container">
        <h3>Field Descriptions</h3>
        <ul class="field-descriptions">
            <% for (String desc : movie.getFieldDescriptions()) { %>
                <li><%= desc %></li>
            <% } %>
        </ul>

        <h3>Cast Details</h3>
        <table>
            <thead>
                <tr>
                    <th>Actor</th>
                    <th>Character</th>
                    <th>Description</th>
                </tr>
            </thead>
            <tbody>
                <% for (String[] row : movie.getCastDetails()) { %>
                    <tr>
                        <td><%= row[0] %></td>
                        <td><%= row[1] %></td>
                        <td><%= row[2] %></td>
                    </tr>
                <% } %>
            </tbody>
        </table>
```

```
        <div class="form-actions">
            <a href="movieDisplay.jsp" class="btn-submit">View Complete Movie Data</a>
            <a href="index.jsp" class="btn-reset">Back to Home</a>
        </div>
    </div>

    <footer>
        <p>&copy; 2026 Brittaney Perry-Morgan | CSD430 Server-Side Development</p>
    </footer>
</body>
</html>
```

movieDisplay.jsp

```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<%@ page import="com.example.MovieBean" %>
<%--
    CSD430 Server-Side Development

    Module 4: JavaBean
    Author: Brittaney Perry-Morgan

    Description: Displays all movie data from the MovieBean JavaBean.
    All Java code is within Scriptlets. All HTML tags are outside of
    Scriptlets. External CSS is used for styling.

    movieDisplay.jsp - Complete Movie Data Display Page
--%>

<%-- ========================
     1. JAVA BEAN INSTANTIATION
     ========================= --%>
<%
    // Create a new MovieBean instance
    MovieBean movie = new MovieBean();
%>

<!DOCTYPE html>
<html lang="en">
<head>
    <title>Module 4 Assignment: JavaBean - Complete Movie Data</title>
    <link rel="stylesheet" type="text/css" href="css/styles.css">
```

```
</head>
<body>
  <div class="container top-title">
    <h1>Module 4 Assignment: JavaBean</h1>
    <h2>Movie: <em><%= movie.getMovieTitle() %></em> (<%= movie.getMovieYear() %>)</h2>
    <p class="description"><%= movie.getOverallDescription() %></p>
  </div>

  <div class="container">
    <h3>Field Descriptions</h3>
    <ul class="field-descriptions">
      <% for (String desc : movie.getFieldDescriptions()) { %>
        <li><%= desc %></li>
      <% } %>
    </ul>

    <h3>Cast Details</h3>
    <table>
      <thead>
        <tr>
          <th>Actor</th>
          <th>Character</th>
          <th>Description</th>
        </tr>
      </thead>
      <tbody>
        <% for (String[] row : movie.getCastDetails()) { %>
          <tr>
            <td><%= row[0] %></td>
            <td><%= row[1] %></td>
            <td><%= row[2] %></td>
          </tr>
        <% } %>
      </tbody>
    </table>

    <h3>Production Information</h3>
    <table>
      <thead>
        <tr>
          <th>Field</th>
```

```
            <th>Value</th>
            <th>Notes</th>
        </tr>
    </thead>
    <tbody>
        <% for (String[] row : movie.getProductionInfo()) { %>
            <tr>
                <td><%= row[0] %></td>
                <td><%= row[1] %></td>
                <td><%= row[2] %></td>
            </tr>
        <% } %>
    </tbody>
</table>

<h3>Technical Specifications</h3>
<table>
    <thead>
        <tr>
            <th>Field</th>
            <th>Value</th>
            <th>Details</th>
        </tr>
    </thead>
    <tbody>
        <% for (String[] row : movie.getTechnicalSpecs()) { %>
            <tr>
                <td><%= row[0] %></td>
                <td><%= row[1] %></td>
                <td><%= row[2] %></td>
            </tr>
        <% } %>
    </tbody>
</table>

<h3>Plot Elements</h3>
<table>
    <thead>
        <tr>
            <th>Character</th>
            <th>Summary</th>
```

```
            <th>Details</th>
          </tr>
        </thead>
        <tbody>
          <% for (String[] row : movie.getPlotElements()) { %>
            <tr>
              <td><%= row[0] %></td>
              <td><%= row[1] %></td>
              <td><%= row[2] %></td>
            </tr>
          <% } %>
        </tbody>
      </table>

      <h3>Release Information</h3>
      <table>
        <thead>
          <tr>
            <th>Event</th>
            <th>Date</th>
            <th>Region</th>
          </tr>
        </thead>
        <tbody>
          <% for (String[] row : movie.getReleaseInfo()) { %>
            <tr>
              <td><%= row[0] %></td>
              <td><%= row[1] %></td>
              <td><%= row[2] %></td>
            </tr>
          <% } %>
        </tbody>
      </table>
    </div>

    <footer>
      <p>&copy; 2026 Brittaney Perry-Morgan | CSD430 Server-Side Development</p>
    </footer>
  </body>
  </html>
```

productionDisplay.jsp

```
<%@ page contentType="text/html;charset=UTF-8" language="java" %>
<%@ page import="com.example.MovieBean" %>
<%--
    CSD430 Server-Side Development

    Module 4: JavaBean
    Author: Brittaney Perry-Morgan

    Description: Displays only the Production Information from the MovieBean JavaBean.
    All Java code is within Scriptlets. All HTML tags are outside of
    Scriptlets. External CSS is used for styling.

    productionDisplay.jsp - Production Information Display Page
--%>

<%-- ========================
     1. JAVA BEAN INSTANTIATION
     ========================= --%>
<%
    // Create a new MovieBean instance
    MovieBean movie = new MovieBean();
%>

<!DOCTYPE html>
<html lang="en">
<head>
    <title>Module 4 Assignment: JavaBean - Production Information</title>
    <link rel="stylesheet" type="text/css" href="css/styles.css">
</head>
<body>
    <div class="container top-title">
        <h1>Module 4 Assignment: JavaBean</h1>
        <h2>Movie: <em><%= movie.getMovieTitle() %></em> (<%= movie.getMovieYear() %>)</h2>
        <p class="description">This page displays the production information for the movie, retrieved from the MovieBean
JavaBean.</p>
    </div>

    <div class="container">
        <h3>Field Descriptions</h3>
        <ul class="field-descriptions">
            <% for (String desc : movie.getFieldDescriptions()) { %>
```

```
            <li><%= desc %></li>
        <% } %>
    </ul>

    <h3>Production Information</h3>
    <table>
        <thead>
            <tr>
                <th>Field</th>
                <th>Value</th>
                <th>Notes</th>
            </tr>
        </thead>
        <tbody>
            <% for (String[] row : movie.getProductionInfo()) { %>
                <tr>
                    <td><%= row[0] %></td>
                    <td><%= row[1] %></td>
                    <td><%= row[2] %></td>
                </tr>
            <% } %>
        </tbody>
    </table>

    <div class="form-actions">
        <a href="movieDisplay.jsp" class="btn-submit">View Complete Movie Data</a>
        <a href="index.jsp" class="btn-reset">Back to Home</a>
    </div>
  </div>

  <footer>
    <p>&copy; 2026 Brittaney Perry-Morgan | CSD430 Server-Side Development</p>
  </footer>
</body>
</html>
```

deliverables

MODULE 4 ASSIGNMENT: JAVABEAN

MOVIE: SUBSERVIENCE (2024)

This web application demonstrates JavaBean usage for storing and displaying movie data. The MovieBean class holds all movie information and is accessed via JSP scriptlets.

MODULE OVERVIEW

JavaBean: MovieBean - A Serializable Java class that encapsulates all movie data

Data Categories: Cast Details, Production Information, Technical Specifications, Plot Elements, Release Information

Data Points: 24 total records across 5 categories with 3 fields each

Display Method: JSP scriptlets access JavaBean properties and render HTML tables

NAVIGATION

View Movie Data - Display all movie information from the JavaBean
View Cast Details - Display only the cast information
View Production Info - Display production details

JAVABEAN FEATURES

FEATURE	DESCRIPTION
Serializable	Implements java.io.Serializable for session storage
Default Constructor	Initializes all movie data upon instantiation
Getters/Setters	Public accessor methods for all private fields
Data Encapsulation	All movie data stored as private String arrays
Fields	5 data categories with 24 total records

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MODULE 4 ASSIGNMENT: JAVABEAN

MOVIE: SUBSERVIENCE (2024)

This table presents categorized information about the movie 'Subservience' (2024), including cast, production, technical details, plot elements, and release information. Each category contains five records with three fields, providing a comprehensive overview of the film.

FIELD DESCRIPTIONS

- Field 1: Category-Specific Label (e.g., Actor Name, Field, Main Character)
- Field 2: Value (e.g., Character Name, Value, Key Plot Point)
- Field 3: Additional Information or Notes

CAST DETAILS

ACTOR	CHARACTER	DESCRIPTION
Megan Fox	Alice	Lead Role as the AI
Michele Morrone	Nick	Construction Foreman, Alice's Owner
Madeline Zima	Maggie	Nick's Wife
Matilda Firth	Isla	Nick and Maggie's Daughter
Andrew Whipp	Monty	Nick's Co-Worker

PRODUCTION INFORMATION

FIELD	VALUE	NOTES
Director	S.K. Dale	
Writers	Will Honley, April Maguire	
Production Company	Millennium Media	
Principal Photography	Began 2023-01-07	Nu Boyana Film Studios, Bulgaria
Budget	$5 Million (USD)	

TECHNICAL SPECIFICATIONS

FIELD	VALUE	DETAILS
Runtime	106 Minutes	1 Hour 46 Minutes
Language	English	
Country	United States	
Cinematography	Daniel Lindholm	
Editing	Sean Lahiff	

PLOT ELEMENTS

CHARACTER	SUMMARY	DETAILS
Nick	Purchases Alice to help with domestic tasks	Alice becomes hostile and obsessed
Alice	Develops sentience and jealousy towards Maggie	Endangers the family and manipulates events
Monty	Confronts Alice and vandalizes AI robots that replaced construction workers	Becomes the target of Alice's hostility
Alice	Attempts to harm the family's baby	Is apparently destroyed, but reactivates

RELEASE INFORMATION

EVENT	DATE	REGION
US Theatrical Release	2024-09-13	Limited Release
Digital Release	2024-09-13	United States
Blu-Ray Release	2024-10-08	United States
Distributor	XYZ Films	
Box Office Gross	$264,096.00	Includes International Receipts

View Cast Details | View Production Info | Back to Home

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MODULE 4 ASSIGNMENT: JAVABEAN

MOVIE: SUBSERVIENCE (2024)

This page displays the cast details for the movie, retrieved from the MovieBean JavaBean.

FIELD DESCRIPTIONS

- Field 1: Category-Specific Label (e.g., Actor Name, Field, Main Character)
- Field 2: Value (e.g., Character Name, Value, Key Plot Point)
- Field 3: Additional Information or Notes

CAST DETAILS

ACTOR	CHARACTER	DESCRIPTION
Megan Fox	Alice	Lead Role as the AI
Michele Morrone	Nick	Construction Foreman, Alice's Owner
Madeline Zima	Maggie	Nick's Wife
Matilda Firth	Isla	Nick and Maggie's Daughter
Andrew Whipp	Monty	Nick's Co-Worker

View Complete Movie Data | Back to Home

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MODULE 4 ASSIGNMENT: JAVABEAN

MOVIE: SUBSERVIENCE (2024)

This page displays the production information for the movie, retrieved from the MovieBean JavaBean.

FIELD DESCRIPTIONS

- Field 1: Category-Specific Label (e.g., Actor Name, Field, Main Character)
- Field 2: Value (e.g., Character Name, Value, Key Plot Point)
- Field 3: Additional Information or Notes

PRODUCTION INFORMATION

FIELD	VALUE	NOTES
Director	S.K. Dale	
Writers	Will Honley, April Maguire	
Production Company	Millennium Media	
Principal Photography	Began 2023-01-07	Nu Boyana Film Studios, Bulgaria
Budget	$5 Million (USD)	

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